Institute/ School Name	School of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Program Name	Bachelor of Engineering (Computer Science & Engineering): B.E (CSE)		
Course Code	25CS030	Course Name	Advanced Web Technologies
L-T-P (Per Week)	3-0-4	Course Credits	5
Academic Year	2025-26	Semester/Batch	6th /2023-2027
Course Coordinator	Dr. Ashutosh Kumar Dubey		

1. **Course Outline:**
   This course offers a comprehensive learning path in full-stack web development, starting with JavaScript and HTML5 Canvas, followed by in-depth TypeScript and React with TypeScript for scalable front-end applications. It covers state management, API integration, and performance optimization, and provides backend experience with Node.js, REST, GraphQL, JWT-based security, and real-time applications using WebSockets and Zego Cloud.

2. **Programme Outcomes (POs):**

At the end of the programme, students will be able to:	
PO 1	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3. Course Learning Outcomes (CLO):

Student should be able to

CLO1: Develop strong analytical skills to analyze modern web application requirements and explain the role of strongly typed languages, component-based architectures, and client–server communication models in building scalable front-end and back-end systems.
CLO2: Apply programming skills in TypeScript and React to develop reusable, well-structured, and type-safe user interface components with effective state management, routing, form handling, and performance optimization techniques, thereby enhancing employability skills.
CLO3: Demonstrate practical development skills to design secure and maintainable backend services using Node.js by implementing RESTful and GraphQL APIs, incorporating validation, pagination, filtering, authentication, authorization, and role-based access control mechanisms.
CLO4: Strengthen professional employability skills by integrating real-time communication and media features through applications using WebSockets, chat systems, and video conferencing capabilities while ensuring reliability, scalability, and security.
CLO5: Evaluate and refactor full-stack applications by applying SOLID principles, architectural patterns, and best practices to improve code quality, performance, and long-term maintainability.

4. CLO-PO Mapping Matrix:

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1	-	H	H	-	H	-	-	-	-	-	-	H
CLO2	-	M	H	L	H	-	-	-	-	-	-	M
CLO3	-	-	H	L	H	-	-	-	-	-	-	M
CLO4	-	-	H	L	H	-	-	-	-	H	-	M
CLO5	-	-	-	-	H	-	-	-	M	H	H	M

5. ERISE Grid Mapping:

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	1
Research/Innovation	1
Skills	5
Employability	5

6. Recommended Books (Reference Books/Text Books):

B01: Cavalcante, D. TypeScript crash course: A hands-on guide to building safer and more reliable web applications (1st Edition). BPB Publications.
B02: Accomazzo, A. Fullstack React: The complete guide to ReactJS and friends (1st Edition). Fullstack.io.
B03: Lyon, W. Full stack GraphQL applications: With React, Node.js (1st Edition). Manning Publications.

7. Other readings and relevant websites:

	Link of Journals, Magazines, Websites and Research Papers
R1	https://fullstackopen.com/
R2	https://frontendmasters.com/courses/
R3	https://react.dev/learn/typescript
R4	https://www.tutorialspoint.com/redis/index.htm
R5	https://dev.to/therickedge/how-to-deploy-your-website-using-vercel-4499
	Link of Audio-Video resources
V1	https://www.youtube.com/watch?v=_9mTJ84uL1Q
V2	https://www.youtube.com/watch?v=Hbt56gFj998
V3	https://www.youtube.com/watch?v=oIsf9zE-TRI

* Resources uploaded on ERP system is accessible to all the students registered for the course.

8. Recommended Tools and Platforms:

- Visual Studio Code, React, Tailwind
- Code Quotient (online platform-https://codequotient.com)

9. Course Plan:

Lecture Number	Topics	Weightage in ETE (%)	Instructional Resources
1-3	Understanding Canvas	15	R1, V1
4-8	Practical: Mario Game using JavaScript		R2, V2
9-11	Typescript Introduction-working with Vs code and configuring it for typescript, Intro of Typescript, Basics of Typescript: variables and datatypes-tuples, union, array		R3, B01
12-13	Objects, Enum, any, unknown, never		R3, B01
14-16	Functions - Functions in context of typescript, Function, Overloading,		R3, B01, V3
17-19	Classes- Classes, use and syntax, Constructors, constructor overloading	25	R3, B01, V3
20-23	Typescript inheritance, interface, module export and import		R3, B01
24-26	Assignment on Typescript		R3, B01
27-29	Revisit React, TypeScript with React		R4, B02
30-32	Function & Class components with props		R4, B03
33-36	State Management: Local vs global state	15	R4, B02
37-39	React: Routing & Navigation		R4, B02
40-42	Forms & validation: Controlled vs uncontrolled forms, react Hook form		R4, B02
43-46	Schema validation (Yup or Zod Validation Libraries)		R4, B02
47-48	State Management: Local vs global state,		R4, B03
49-51	Redux toolkit architecture, async thunks, normalized state		R4, B02
52-55	Redux Toolkit: API caching, Auto re-fetching, optimistic updates, error handling	15	R4, B02, V1
56-60	Role Based Access Control, API Security and best practices		R4, B02, V2
61-65	Performance Optimization in React: Re-render issues, useMemo, useCallback, lazy loading		R4, B01
66-67	Reusable Components		R4, B02
68-70	UI patterns & MVC	10	R4, B02
71-75	Client-Server Architecture, SPA vs MPA, Rest vs GraphQL		R5, B03
76-79	Authentication using JWT: Access vs Refresh Token, Token lifecycle, Auth middleware		R5, B03, V3
80-84	Nodejs, Rest API Design: status codes, pagination & filtering, Validation middleware		R5, B02
85-87	GraphQL with Node.js: Schema, Resolvers, Queries & mutations	20	R5, B02
88-91	Web Socket: Introduction to web sockets, Socket.io		R5, B02
82-94	Creating a Chatting Application		R5, B02
95-98	Implementing Google meet like application with video capturing using Zegocloud		R5, B03
99-101	SOLID Principles		R5, B02, V3
102-105	Assignments on Full Stack		R5, B02

10. Industry Interventions:

- The students will receive resources on the LMS system, and their progress will be tracked on the platform based on their submissions (number of submissions, errors encountered, and optimized solutions).

11. Action plan for different types of learners

Slow Learners	Average Learners	Advanced Learners
Remedial Classes	Workshops / Practice Assignment	Advance practice problem

12. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Internal Component	Sessional Tests	02*	40%	Online**
External Component	End Term Examination (ETE)	01	60%	Online**
Total		**100%**		

* Out of the two, the best 01 will be considered.
** Proctored examination will be conducted on Testpad platform.

13. Details of Evaluation Components:

Evaluation Component	Description	Syllabus Covered (%)	Timeline of Examination	Weightage (%)
Internal Component 2	ST 01	Upto 40% (Lectures 1-42)	Week 7	40%
	ST 02	41% - 80% (Lectures 43-84)	Week 12	
External Component	End Term Examination*	100%	As Notified by the Exam Cell	60%
Total				100%

* Minimum 75% attendance is required to become eligible for appearing in the End Term Examination

14. Format of Evaluation Components:

Type of Assessment	Total Marks	1 Mark MCQ	2 Marks MCQ	5 Marks Coding Question	10 Marks Coding Question
Sessional Tests	40	10	5	2	1
End Term Examination	60	10	10	2	2

15. Revision (if any):

Academic Yearof Previous Version	2024-2025	Percentage of Revision	13%
Topics:			
Added: • Understanding Canvas • React Routing and Navigation, Forms & validation: Controlled vs uncontrolled forms, react Hook Form, Schema validation (Yup or Zod Validation Libraries) • Redux toolkit architecture, async thunks, normalized state, Redux Toolkit: API caching, Auto re-fetching, optimistic updates, error handling • GraphQL with Node.js: Schema, Resolvers, Queries & mutations • SOLID Principles			

16. This Document is approved by:

Designation	Name	Signature
Prepared by Course Coordinator	Dr. Ashutosh Kumar Dubey	
Verified by Assistant Dean	Dr. Ashutosh Kumar Dubey | Dr. Ravita Chahar	
Date	January 10, 2026	